P_311 — Knowledge-Driven IoT Fault Diagnosis Assistant

An End-to-End Industrial Predictive Maintenance System with Gradient Boosting ML, FAISS RAG, and a Local LLM Shop-Floor Copilot

Learner & Mentor Credentials

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Key Innovations & Deliverables

- Multi-Machine MQTT Telemetry (3 Industrial Machine Cell)
- Gradient Boosting Classifier (0.9696 ROC-AUC, 0.40 Threshold)
- FAISS Dense Vector RAG (all-MiniLM-L6-v2 Embeddings)
- Local LLM Copilot Briefing (100% Zero-Hallucination Grounding)
- Streamlit Real-Time Dashboard with Plotly Gauges, XAI & RUL

Industrial Problem Statement & The Need for Knowledge AI

1. Static Threshold Failures

The Problem: Traditional SCADA/IoT alarms fire only after extreme thresholds (e.g., Temp > 80°C) are breached.

Compound Risk: In modern manufacturing, subtle correlations (high torque + high tool wear + mild heat) cause catastrophic breakdowns while individual variables appear 'nominal'.

Consequence: Unplanned factory stoppages, damaged workpieces, and millions lost in production downtime.

2. The 'Black-Box' Alarm Gap

The Problem: Standard ML alerts emit a bare probability number (e.g., 'Risk: 84%') without operational clarity.

Technician Dilemma: Shop-floor operators do not know which physical sensor is the root cause.

Manual Overload: Technicians must hunt through 500-page OEM binder manuals to find proper torque specs or tool replacement SOPs during an active emergency.

3. The P_311 Solution

Integrated Vision: Closed-loop synthesis of ML prediction, physical rule isolation, vector RAG, and an LLM Copilot.

Explainable AI (XAI): Live sensor attribution bar charts show exactly why failure risk is elevated.

Instant OEM Guidance: FAISS vector database retrieves certified procedures in <5ms; local copilot briefs the technician instantly.

End-to-End System Architecture & Data Pipeline

1. Edge Sensing

Physical Sensors / Fleet Simulator

- CNC Milling (M)
- Heavy Lathe (H)
- High-Speed Grinder (L)
- Speed (RPM)
- Torque (Nm)
- Cutting Temps (K)
- Tool Wear (min)

2. Transport

Mosquitto MQTT :1883

- Topic: factory/+/sensors
- Asynchronous I/O
- JSON Over TCP
- Sub-10ms delivery
- Decoupled Fleet Scalability

3. Ingestion & ML

ai4i_ml_receiver.py

- Gradient Boosting Model
- 8-Feature Leak-Free Input
- Calibrated Threshold 0.40
- 0.9696 ROC-AUC Score
- Physical Rule Engine
- Severity Grader

4. RAG & Copilot

FAISS & Local LLM

- Sentence Transformers (384-d)
- FAISS Vector Index
- OEM SOP Retrieval
- Stage 6 LLM Copilot
- Grounded Fallback Engine
- Zero Hallucination

5. UI & DB

Dashboard :8501

- Plotly Gauges
- XAI Bar Chart
- RUL Countdown
- SQLite Audit Log
- CSV Exporter

Multi-Machine IoT Fleet Simulation & Physics Engine

3-Machine Factory Cell Telemetry

machine1: CNC Milling Center (Type M)
Balances medium load with high precision; common in automotive parts milling.

machine2: Heavy Duty Lathe (Type H)
High-torque turning center; handles large workpieces with high mechanical stress.

machine3: High Speed Grinder (Type L)
Lightweight, ultra-high spindle speeds; sensitive to vibration and heat dissipation.

Physical Simulation Parameters & Telemetry

Sampling Rate: Every 2.0 seconds per machine via MQTT

Air Temperature: 295.0 K to 304.0 K (Factory ambient)

Process Temperature: 305.0 K to 314.0 K (Cutting zone temp)

Rotational Speed: 1,100 to 2,800 RPM (Inverted torque relation)

Spindle Torque: 15.0 to 75.0 Nm (Mechanical load)

Tool Wear: 0 to 240 minutes (Degradation accumulator)

Fault Injection: 80% nominal operating points, 20% realistic failure patterns (thermal dissipation, overstrain, stall).

Stage 4: Gradient Boosting ML Failure Prediction

AI4I 2020 Dataset & Zero Leakage

Dataset Source: UCI Machine Learning Repository (10,000 real-world synthetic milling records).

Class Imbalance: Only 3.39% failures (339 positive cases out of 10,000). Standard accuracy is deceptive.

No Target Leakage: Crucial engineering decision: Columns 'TWF', 'HDF', 'PWF', 'OSF', 'RNF' directly encode the failure mode. Including them causes artificial 100% test accuracy. We strictly exclude them and train only on raw physics variables.

Feature Vector (8): Air Temp, Process Temp, Speed, Torque, Tool Wear, Type_H, Type_L, Type_M.

Model Performance & Operating Threshold

ROC-AUC Score: 0.9696 (Exceptional discriminatory ranking power).

Why Gradient Boosting? Decision trees naturally handle non-linear tabular interactions with sub-millisecond CPU inference.

Calibrated Threshold (0.40): In predictive maintenance, a False Negative (missed breakdown) costs 100x more than a False Positive (early check).

Tuning Result: Calibrating threshold from 0.50 down to 0.40 boosted minority failure recall from 58.8% to 72.1%, with strong 84.5% precision.

Physical Sensor Rule Engine & Severity Grading

Physical Condition Thresholds (diagnosis.py)

Elevated Air Temp: >= 301.0 K (Factory ambient heat spike)
High Process Temp: >= 312.0 K (Cutting zone thermal dissipation stress)
Low Rotational Speed: < 1,300 RPM (Spindle motor stall danger)
High Rotational Speed: > 1,650 RPM (Over-speeding / runaway load)
High Mechanical Torque: >= 55.0 Nm (Excessive motor strain or binding)
Critical Tool Wear: >= 150 min (Maximum certified cutting limit)
Increasing Tool Wear: >= 100 min (Preventive inspection recommended)

4-Tier Risk Severity Classifier

CRITICAL (Prob >= 75%): Immediate machine hazard. Directive: E-STOP or isolate machine immediately. Disclose tool & spindle.

HIGH (Prob >= 40%): Operating threshold breached. Directive: Schedule urgent maintenance within current shift.

MEDIUM (Prob >= 20%): Early signs of mechanical or thermal degradation. Directive: Continue cycle with heightened monitoring.

LOW (Prob < 20%): Nominal operating envelope. Directive: Continue scheduled production cycle without interruption.

Stage 5: FAISS Semantic Vector RAG Architecture

Why Vector RAG Over Keyword Search?

Keyword Search Flaw: Exact string matching fails when technician terminology differs (e.g. searching 'spindle overload' misses an SOP titled 'drive resistance and torque spike').

Dense Embeddings: We utilize 'sentence-transformers/all-MiniLM-L6-v2' to map text chunks into 384-dimensional semantic space.

Cosine Similarity: Calculates semantic conceptual distance regardless of specific vocabulary.

Zero Hallucination: RAG forces maintenance recommendations to be strictly grounded in verified OEM manuals.

FAISS Indexing & Retrieval Performance

FAISS Index Type: IndexFlatIP (Inner Product on L2-normalized vectors).

Knowledge Corpus: maintenance_knowledge.txt (Categorized SOPs for High Torque, Tool Wear, Thermal Dissipation, and Spindle Stall).

Query Latency: Under 5.0 milliseconds on edge CPU.

Dynamic Updates: When machine SOPs update, the knowledge base is re-indexed in seconds without retraining expensive ML or LLM weights.

Stage 6: Local LLM Shop-Floor Copilot

Multi-Backend Architecture (llm_assistant.py)

Role: Acts as an intelligent junior reliability engineer, turning raw telemetry numbers and dense technical manuals into an actionable briefing.

Primary Engine: Local Ollama HTTP API (llama3.2:1b, phi3, mistral). Operates 100% on-premise without cloud data leakage.

Grounded Fallback Engine: A 3.0-second timeout ensures zero shop-floor lag. If Ollama is offline or uninstalled, an internal deterministic grounded synthesizer takes over instantly with zero latency.

Structured 3-Part Briefing Output

1. Situation Summary: Plain-English status statement including live failure risk percentage, spindle RPM, torque, and temperatures.

2. Root-Cause Assessment: Pinpoints which physical factors breached nominal limits (e.g. 'High cutting torque at 71.6 Nm with tool wear at 204 min').

3. Immediate Action Directives: Step-by-step checklist directly extracted from the OEM manual (e.g. 'Check guideways for binding, inspect insert for chipping, replace tool').

Explainable AI (XAI) & Remaining Useful Life (RUL)

Explainable AI (XAI) Feature Attribution

The Black-Box Dilemma: Plant managers distrust alarms when they cannot see what the algorithm evaluated.

Real-Time Attribution Bar: Computes relative normalized contribution weights across active sensors: Torque, Tool Wear, Rotational Speed, and Temperature Delta.

Shop-Floor Value: Technicians immediately see whether an 80% failure risk is driven by mechanical spindle resistance (Torque) or abrasive degradation (Tool Wear), saving critical diagnostic minutes.

Remaining Useful Life (RUL) Tracking

Beyond Binary Alarms: Traditional alarms only trigger when broken. RUL enables predictive replacement before failure happens.

Tool Replacement Boundary: AI4I standard specifies 150.0 minutes of certified cutting life.

RUL Formulation: RUL = max(0, 150 - Tool_Wear).

Visual Indicators: Color-coded countdown badges: Green (>50 min), Amber (15-50 min), and Flashing Red (<15 min / Critical Expiry).

Bill of Materials (BOM) & Edge Hardware Architecture

Industrial Edge Deployment Hardware (Per Machine Cell)

ESP32 DevKit v1 (38 Pins): Central IoT edge node running FreeRTOS & MQTT client (₹450)

MAX6675 + K-Type Thermocouple: Cutting process temperature measurement up to 600°C (₹280)

SCT-013 Non-Invasive Current Clamp: Spindle current draw to derive motor torque without cutting 415V wires (₹320)

E6B2-CWZ6C Optical Rotary Encoder: Spindle speed (RPM) pulse accumulator via hardware interrupts (₹650)

0.96" I2C OLED Display (SSD1306): Local machine-side IP and health state display (₹180)

LM2596 Buck Converter + IP65 Enclosure: Factory 24V-to-5V power conversion & splash-proof housing (₹370)

Total Hardware Cost: ₹2,935 INR / Machine Cell (Extremely cost-effective for retrofitting legacy factory machinery!)

System Verification, Test Benchmarks & Resilience

Test Matrix (10 / 10 Tests Passed)

TC-01 MQTT Connection & Ping: PASS (Mosquitto 1883)
TC-02 Multi-Machine Multiplexing: PASS (Topics factory/+/sensors)
TC-03 ML Pipeline Inference Latency: PASS (<2 ms on CPU)
TC-04 Target Leakage Exclusion: PASS (TWF/HDF/PWF excluded)
TC-05 Calibrated Threshold Recall: PASS (72.1% Recall at 0.40)
TC-06 Physical Rule Engine Checks: PASS (Multi-condition isolation)
TC-07 FAISS Vector Retrieval: PASS (<5 ms Cosine Search)
TC-08 Copilot Grounded Fallback: PASS (Zero latency fallback)
TC-09 SQLite & JSON Cache Sync: PASS (Atomic persistence)
TC-10 Streamlit UI Gauge Rendering: PASS (HTTP 200 OK)

Engineering Edge-Case Resilience

Subprocess Pipe Deadlock Fix: Resolved an OS pipe buffer overflow where unread stdout blocked background processes. Rerouted all logs to persistent log files in 'logs/'.

Console UTF-8 Encoding Fix: Resolved Windows cp1252 console UnicodeEncodeErrors by enforcing utf-8 stdout reconfiguration with error replacement.

Continuous Operation: Verified stable continuous multi-process execution with zero memory leaks.

Viva Defense Key Takeaways & Project Impact

Winning Technical Viva Defenses

Q: Why Gradient Boosting over Deep Learning? For tabular physical sensor data, ensemble trees outperform deep neural nets in sample efficiency, interpretability, and edge CPU speed without GPU power requirements.

Q: Why set decision threshold to 0.40 instead of 0.50? In predictive maintenance, the operational cost of an undetected failure (stoppage) is vastly higher than a false alarm. Lowering threshold boosts recall from 58.8% to 72.1%.

Q: Why RAG instead of fine-tuning an LLM? RAG guarantees 0% hallucination by anchoring in OEM texts, allows instant manual updates without retraining, and runs on CPU.

Project Summary & Future Scope

Project Status: 100% complete, fully verified, and production ready.

GitHub Repository: github.com/kushagrasharoya/Knowledge-Driven-IOT-Fault-Diagnosis-Assistant

Live Video Walkthrough: https://youtu.be/co-VyIK7CFI

Commercial Viability: Under ₹3,000 per machine cell, enabling rapid legacy brownfield smart factory modernization.

Future Roadmap: Hardware-in-the-loop edge deployment, acoustic vibration FFT, and automated ERP spare parts replenishment.